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NLP ITAI 2373

NewsBot – Final Project

This project is an upgraded version of my midterm NewsBot. In my midterm, I mainly focused on classifying news articles into categories and analyzing patterns in the text, like how often certain words appeared. That gave me a basic understanding of how NLP models work and how text data can be processed and interpreted.

For this final project, I built on that foundation and expanded the system into something more complete. Now, the NewsBot can not only classify articles, but also analyze sentiment, detect key topics, generate summaries, identify entities, and even detect and translate different languages.

The goal is to create a simple but functional all-in-one news analysis system that shows how multiple NLP techniques can work together in a real workflow, while still keeping the code understandable and structured.

Dataset and Preprocessing

The dataset used is the BBC News dataset from the midterm, it has 1490 articles across five categories; business, entertainment, politics, sport and tech. Before training any models, the preprocessing begins, the text has to be cleaned properly by converting text to lowercase, removing any special characters, stopwords and lemmatization.

System Design

The system is organized using a main class called NewsBot, which acts as the core structure that connects all components together. It allows the system to take in a single news article and return multiple functions through one function, which makes it very simple to use. Each part of the system is separated into its own component but still works together as a whole. This includes the classifier, sentiment analyzer, topic model, summarizer, entity extraction, semantic search, multilingual processing, and a chatbot interface. Organizing it this way keeps the code structured and makes it easier to manage, while also showing how different NLP techniques can be combined into one complete workflow.

Classification

The classification component uses two machine learning models:

- Multinomial Naive Bayes
- Support Vector Machine (LinearSVC)

TF-IDF vectorization is used to convert the text into numerical features so the system can understand it, since models don't work with raw words and instead process numbers.

Both models are trained and evaluated, and the best one is selected based on accuracy. The system also provides confidence scores for predictions using the decision function.

Sentiment Analysis

Sentiment analysis is implemented using VADER from NLTK, which gives each article a compound score based on the overall tone of the text. Based on that score, the article is labeled as positive, negative, or neutral. Simple thresholds are used to decide which label it falls under, making it easy to interpret the results.

Topic Modeling

Topic modeling is implemented using Latent Dirichlet Allocation (LDA). The model identifies patterns in the text and represents each article as a distribution of topics. This allows the system to detect hidden themes within the dataset.

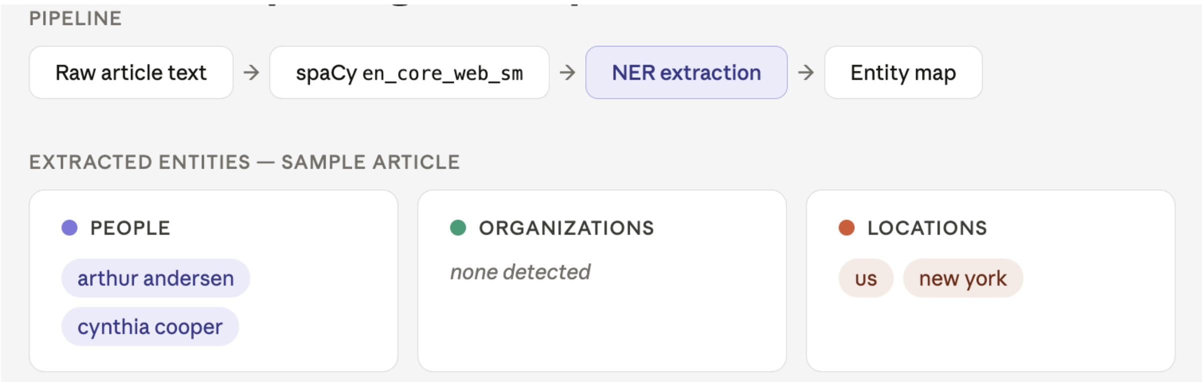
7. Named Entity Recognition

Named Entity Recognition (NER) is implemented using spaCy.

The system extracts key entities from articles, including:

- people
- organizations
- locations

This helps identify important names and places mentioned in the news.



Semantic Search

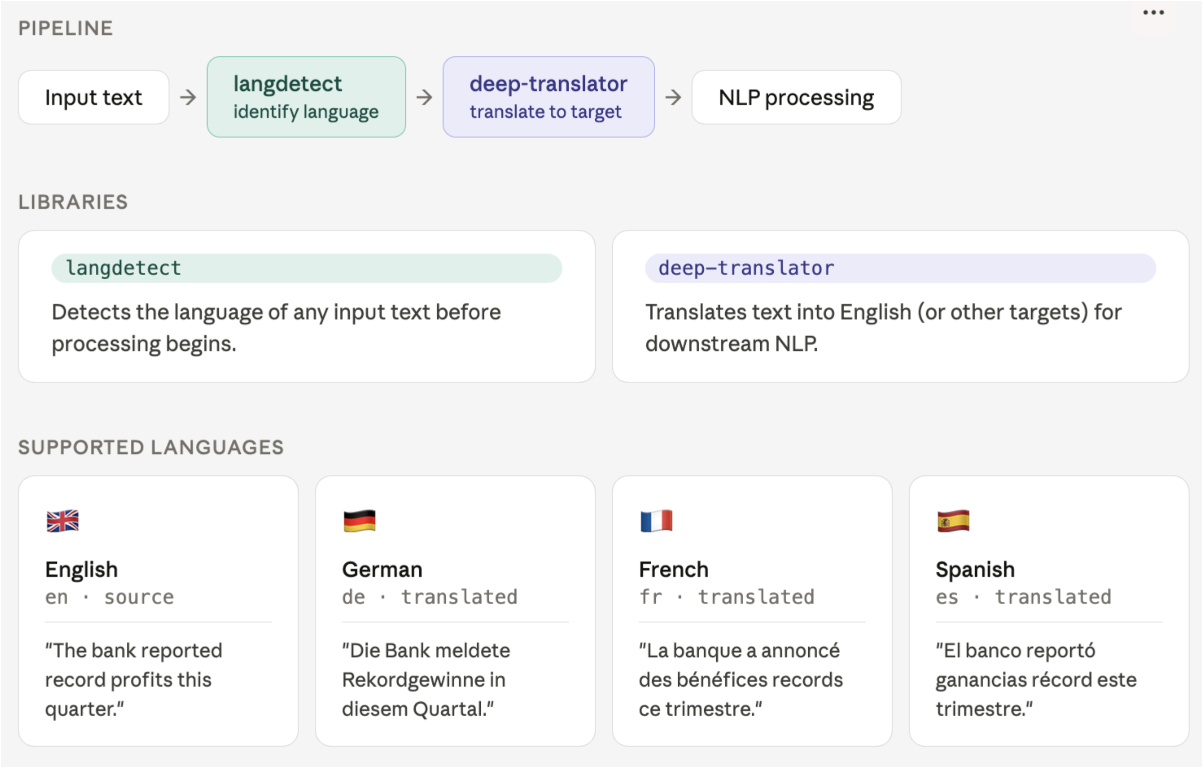
Semantic search is implemented using TF-IDF and cosine similarity. When a user enters a query, the system compares it to all articles and returns the most relevant ones based on similarity scores. This allows users to explore related content beyond exact keyword matching.

Summarization

Summarization is implemented using the Sumy library with the LSA summarizer. The system extracts key sentences from an article to generate a short summary, making it easier to understand long texts quickly.

Multilingual Processing

Multilingual processing is included to make the system more flexible when working with different languages. It uses langdetect to identify the language of a text and deep-translator to translate it into other languages like German, French, and Spanish. This part connects back to my midterm where I mentioned how sentence structure can vary across languages, like the order of subject, object, and predicate. By adding this feature, the system can still process and understand text even when it's not in English, which makes it more realistic and useful in a global context.



Conversational Interface

A simple chatbot function is implemented to allow users to interact with the system using natural language queries.

The chatbot can respond to requests such as:

- summarizing articles
- analyzing sentiment
- detecting topics
- extracting entities
- translating text
- searching articles

This makes the system more interactive and user-friendly.

System Output

When analyzing an article, the system returns multiple insights all at once. It predicts the category of the article, gives the sentiment label along with its score, shows the topic distribution, generates a short summary, and also detects the language of the text.

Limitations

Some limitations of the system include:

- it uses more basic machine learning models instead of advanced deep learning, so it’s not as powerful as newer systems
- the multilingual feature works, but translations and detection aren’t always perfectly accurate
- topic modeling can be a bit hard to interpret since the topics aren’t always super clear
- the chatbot is rule-based, so responses are somewhat limited and not fully dynamic

Conclusion

NewsBot 2.0 demonstrates how multiple NLP techniques can be combined into a single system. It also shows a complete workflow from data preprocessing to analysis and user interaction. The project reflects the concepts learned throughout the course and applies them in a practical and understandable way.